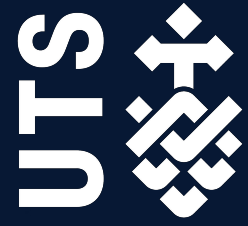


Assignment 2:

Data Analysis Report



36103: Statistical Thinking for Data Science

Submitted: Saturday 9th May, 2026

Group Number: 5

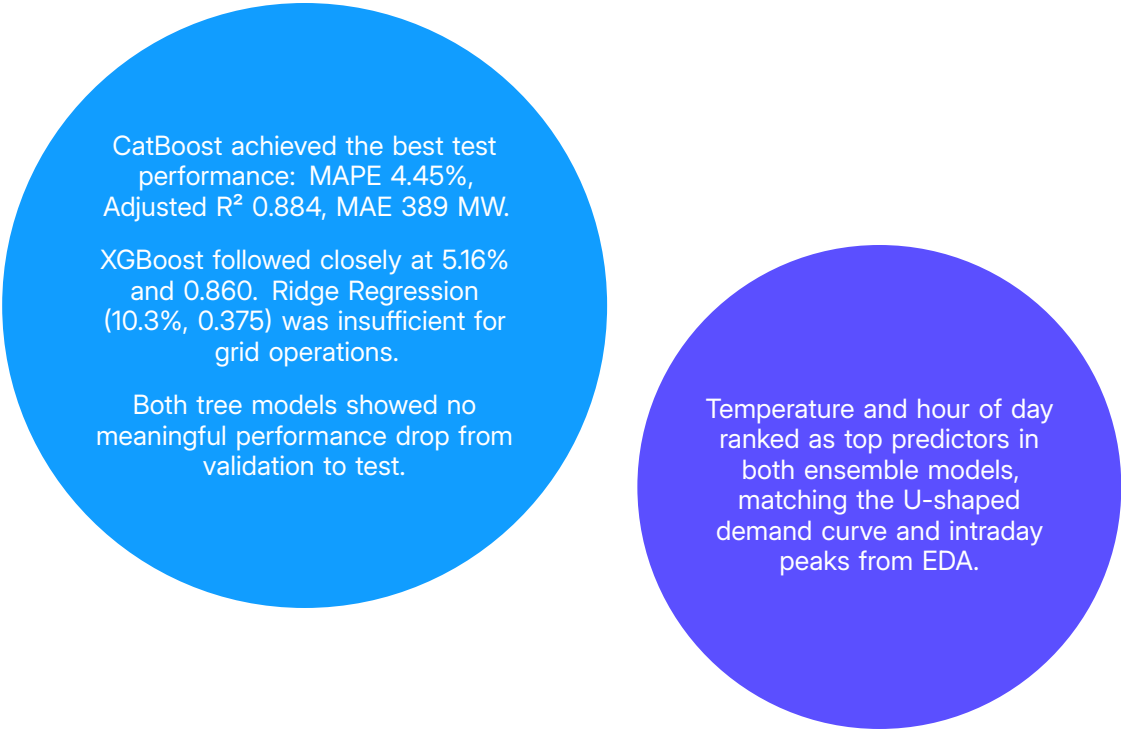
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Executive Summary

Electricity demand in NSW changes with weather, time of day, and season. During peak periods, wholesale prices can reach \$17,500/MWh, so reliable short-term forecasts matter for both cost management and grid stability.

This project used high-frequency NSW data from 2022-2026 to study demand patterns and build forecasting models. Exploratory analysis showed demand is lowest in mild conditions, rises during extreme hot or cold weather, and differs between weekdays and weekends. Rooftop solar reduces midday demand but causes a sharp evening increase.

CatBoost is the recommended model. Its average error of around 389 MW means forecasts should inform decisions, not drive them outright, particularly during extreme weather or supply disruptions.



CatBoost achieved the best test performance: MAPE 4.45%, Adjusted R^2 0.884, MAE 389 MW.

XGBoost followed closely at 5.16% and 0.860. Ridge Regression (10.3%, 0.375) was insufficient for grid operations.

Both tree models showed no meaningful performance drop from validation to test.

Temperature and hour of day ranked as top predictors in both ensemble models, matching the U-shaped demand curve and intraday peaks from EDA.

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Introduction

1 Introduction

1.1 Problem Statement

Electricity supply and demand must balance every 30 minutes. When they don't, grid operators absorb the cost. NSW wholesale prices can reach \$17,500/MWh under the market cap that applied through mid-2025 (AER, 2025c). Accurate short-term demand forecasting keeps dispatch decisions ahead of those spikes.

1.2 Rationale

The National Electricity Market (NEM) serves roughly 11 million customers across eastern Australia, with NSW being the country's largest electricity generating state (AER, 2026). Demand shifts every 30 minutes with temperature, time of day, and day type, and the grid has no meaningful storage to absorb mismatches.

The supply side is also changing. Australia ranks first globally in per capita solar electricity generation (Green.com.au, 2026), with NSW holding the highest installed rooftop capacity of any state at 7.5 GW as of mid-2025 (CEC, 2025). Rooftop solar suppresses midday grid demand but drops off sharply in late afternoon, creating a steep evening ramp that dispatchable generators must cover at short notice. NSW already sees a 34-41% gap between its daily minimum and maximum demand, among the largest swings in the NEM (AER, 2025b).

Prices reflect that volatility. In 2024, NSW average wholesale electricity prices rose 43% year-on-year to \$150/MWh (AER, 2025a). Forecasting errors at that price level carry direct costs for operators, retailers, and consumers.

1.3 Project Aims and Objectives

The aim of this project was to determine whether machine learning models trained on historical half-hourly NSW data can predict grid electricity demand accurately enough to support operational dispatch planning. To support this aim, we asked:

1. Does temperature have a non-linear relationship with NSW grid demand across 2°C bins from 5°C to 43°C?
2. Do mean hourly demand profiles differ between weekdays, weekends, and public holidays in NSW?
3. Can an ensemble regression model predict 30-minute NSW grid demand with a MAPE below 5% on held-out test data?

To answer these questions, we sourced and merged half-hourly AEMO data with weather, rooftop solar, and economic indicators, then ran exploratory analysis to identify key demand drivers. We built and compared Ridge Regression, XGBoost, and CatBoost models on a 60:20:20 chronological split, evaluating each on RMSE, MAE, MAPE, and Adjusted R².

Methodology

2 Methodology

2.1 Models

2.1.1 Linear Regression (Ridge)

2.1.2 Gradient Boosting Trees

2.1.2.1 XGBoost

2.1.2.2 CatBoost

2.2 Grid Search

Hyperparameter optimization is performed via Grid Search with k -fold Cross-Validation. Through exhaustive predefined parameter grids, we identified configurations that minimised generalization error of each model.

2.3 Metrics

Table 1. Model Performance Metrics

Metric	Purpose	Formula
MAE	Average absolute error in MW; operationally interpretable.	$\frac{1}{n} \sum y_i - \hat{y}_i $
MAPE	Scale-independent accuracy across demand magnitudes.	$\frac{1}{n} \sum \left \frac{y_i - \hat{y}_i}{y_i} \right \times 100$
RMSE	Penalises large outliers in peak demand periods.	$\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$
Adjusted R²	Explained variance penalised for predictor count.	$1 - \frac{(1-R^2)(n-1)}{n-p-1}$

Results

3 Results

Conclusion

4 Conclusion

4.1 Conclusion

4.2 Discussion

4.3 Project Limitations and Caveats

4.4 Stakeholder Analysis and Project Outcomes

References

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Appendices

6 Appendices